Bank Marketing Analytics Dashboard

Dashboard Findings and Exploratory Data Analysis Report

Page 1 — Customer Analysis

1. Age-Wise Subscription Distribution

Customers aged 30–40 account for the highest number of subscriptions. Subscription counts decline among both younger and older age groups.

Business implication: The bank could prioritize these age groups for further campaign analysis and planning.

2. Subscription Distribution Across Occupations

Customers working in management, technician, and blue-collar occupations account for the majority of subscriptions. This may be because these groups receive a greater proportion of sales calls.

Business implication: The bank could consider increasing outreach to other professional groups to explore additional subscription opportunities.

3. Education and Occupation Segmentation

Within the management occupation, subscription opportunities may exist among customers with primary or secondary education. The current dashboard indicates that outreach is concentrated more heavily on customers with tertiary education.

Business implication: The bank could evaluate increasing contact with management professionals who have primary or secondary education to determine whether this segment offers additional subscription potential.

4. Marital Status and Loan Profile

Based on the pie chart, married customers without loans appear to be a promising segment for term-deposit subscriptions. More generally, customers without loans appear to have a better likelihood of subscribing.

Business implication: The bank could consider allocating more marketing effort to customers without loans, while comparing subscription rates across customer groups to validate this opportunity.

5. Combined Customer Characteristics

The dashboard indicates that married customers without loans and with lower levels of education account for a notable subscription pattern.

Business implication: The bank could examine marital status, loan profile, and education together to identify customer segments that may respond well to future marketing campaigns.

6. Credit-Default Status and Subscription Behaviour

Customers recorded as credit defaulters generally show lower subscription activity across different age groups, education levels, and marital statuses.

Business implication: The bank could examine credit-default status as a factor in campaign segmentation, while taking the size of this customer group into account.

New Skills and Techniques Learned

• Used DAX `SWITCH` logic to create age ranges from the original age column.

• Used bitwise AND logic to combine personal-loan and housing-loan indicators into a derived column.

Screenshots — Customer Analysis

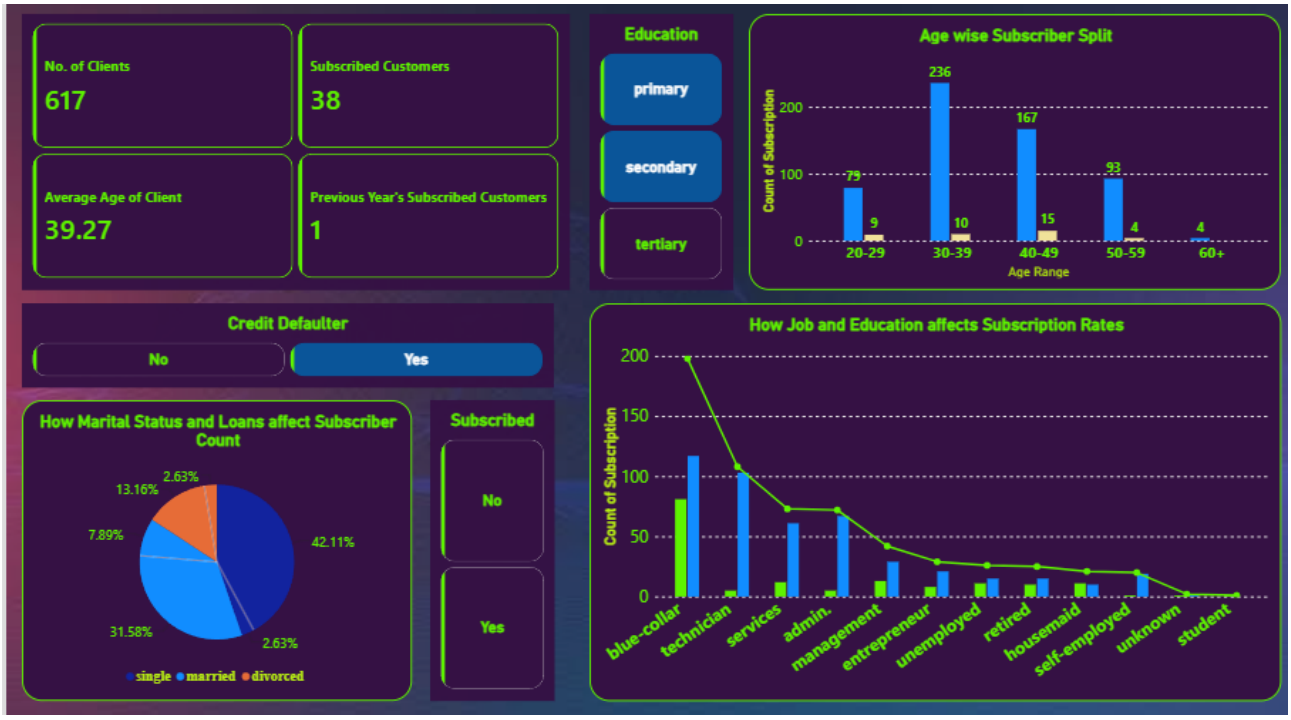
Screenshot 1



Screenshot 2



Screenshot 3



Page 2 — Campaign and Contact Analysis

1. Subscription Distribution by Contact Method

Across both customers who subscribed and those who did not, the highest number of calls was made using cellular contact.

Business implication: The bank could compare subscription rates across contact methods to assess whether cellular contact is not only the most frequently used method but also the most effective.

2. Subscription Rates by Previous Contact History

Approximately 10% of customers contacted for the first time in the current campaign subscribed, compared with approximately 25% of customers who had been contacted during a previous campaign.

Business implication: The bank could investigate whether customers with previous campaign contact are more receptive to future outreach and consider prioritizing follow-up with these customers.

Note: Previous campaign contact history does not necessarily indicate whether someone is a new or existing bank customer.

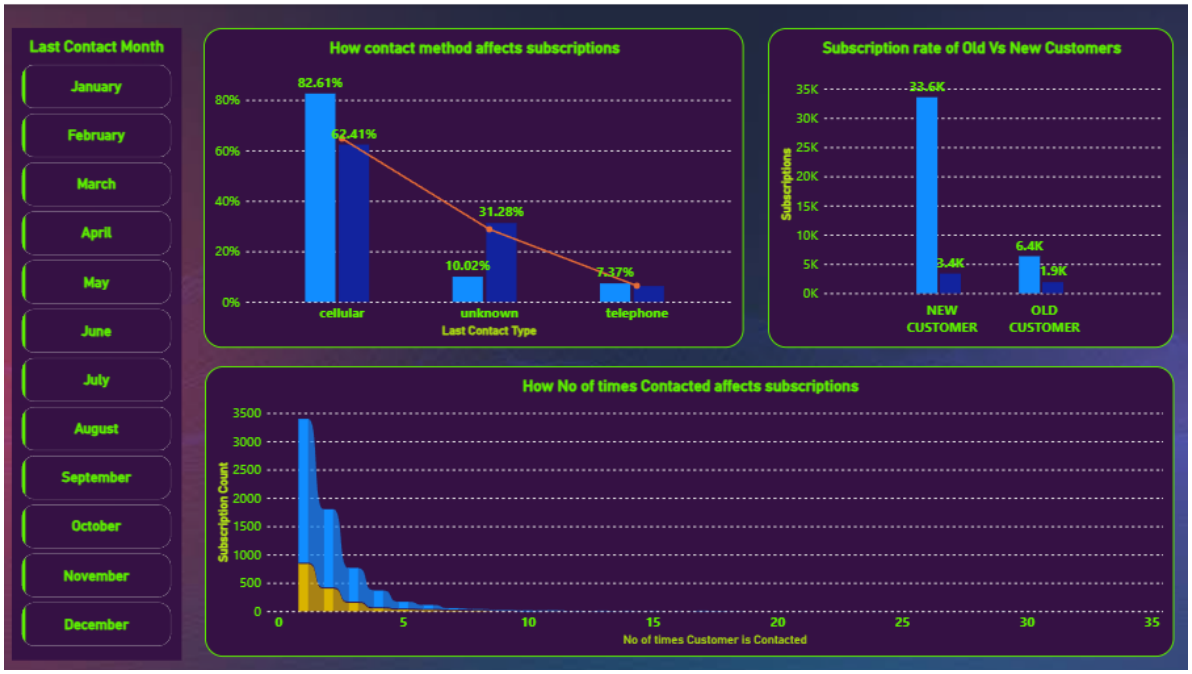
3. Subscription Outcomes by Number of Contacts

Customers contacted fewer than four times during a campaign show the highest probability of subscribing, according to the dashboard. This pattern appears consistent across the two campaign comparisons represented in the analysis.

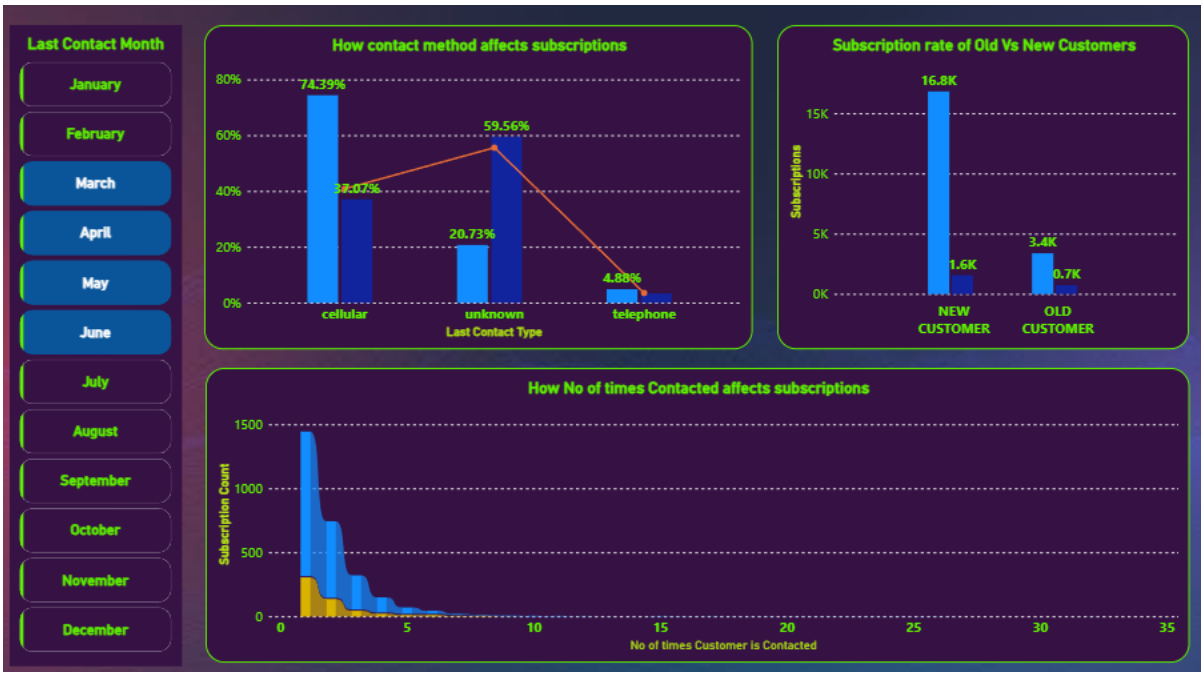
Business implication: The bank could evaluate whether a more focused contact strategy—rather than repeated calls to the same customer—could improve campaign efficiency. The four-contact threshold should be treated as an observed pattern to investigate, rather than a confirmed optimum.

Screenshots — Campaign and Contact Analysis

Screenshot 4



Screenshot 5



Exploratory Data Analysis (EDA) and Data Preparation

The following steps were performed to prepare the dataset and support dashboard development.

1. Exploratory Data Analysis: Performed EDA to examine the dataset, understand its structure, and identify patterns relevant to the dashboard.

2. Column Renaming: Renamed columns to make them easier to identify and interpret during analysis.

3. Range Creation: Created age ranges and call-duration ranges from the original columns using DAX formulas.

4. Calculated Columns and Measures: Created additional columns and measures from existing data to simplify analysis and support dashboard visuals.

5. Date Cleaning: Cleaned the date values and converted them into an appropriate date format for consistent use in the dashboard.